Earth gravity can set up their homes near the inner or outer face of the ring, while those who prefer weaker gravity can live nearer its center. These differences in gravity are due in part to the rotating ring's centrifugal force and in part to the hole's tidal force—or, in Einstein's language, to the curvature of spacetime.

The electric power that heats and lights this ring world is extracted from the black hole: Twenty percent of the hole's mass is in the form of energy that is stored in the tornado-like swirl of space outside but near the horizon.[42] This is 10,000 times more energy than the Sun will radiate as heat and light in its entire lifetime!—and being outside the horizon, it can be extracted. Never mind that the ring world's energy extractor is only 50 percent efficient; it still has a 5000 times greater energy supply than the Sun.

The energy extractor works on the same principle as do some quasars[43]: Your crew have threaded a magnetic field through the hole's horizon and they hold it on the hole, despite its tendency to pop off, by means of giant superconducting coils (Figure P. 6). As the horizon spins, it drags the nearby space into a tornado-like swirl which in turn interacts with the threading magnetic field to form a gigantic electric power generator. The magnetic field lines act as transmission lines for the power. Electric current is driven out of the hole's equator (in the form of electrons flowing inward) and up the magnetic field lines to the ring world. There the current deposits its power. Then it flows out of the ring world on another set of magnetic field lines and down into the hole's north and south poles (in the form of positrons flowing inward). By adjusting the strength of the magnetic field, the world's inhabitants can adjust the power output: weak field and low power in the world's early years; strong field and high power in later years. Gradually as the power is extracted, the hole will slow its spin, but it will take many eons to exhaust the hole's enormous store of spin energy.